

My Programs > Executive Post Graduat... > C4 > MA

≡ Study Plan

< MLOps Capstone - Overview

>

Problem Statement

A heavy-equipment manufacturer is facing significant challenges due to **unplanned machine downtime**, which leads to production delays, increased maintenance costs, and operational inefficiencies.

Each machine continuously generates sensor data reflecting its operational condition. The goal is to use this data to build a system that can **predict whether a machine is operating normally or is likely to fail**, and if a failure is expected, identify the specific failure type.

You are provided with:

- Historical labelled training data representing baseline machine behaviour
- A **stable post-deployment batch** reflecting normal operating conditions
- A **stress batch** collected during heavy-load conditions, representing a potential shift in data distribution

The **train.csv** dataset is used for training and validation, **current.csv** represents stable production data, and **stress.csv** simulates a drifted scenario (valid in schema but different in distribution). This highlights a key MLOps concept: **valid data is not always stable data**.

Your task is to develop a local MLOps workflow that ensures reliable and production-ready predictions. The system should:

- Validate data quality before model usage
- Train and compare multiple models
- Select and tune the best-performing model
- Monitor incoming data for distribution shifts (data drift)
- Explain model predictions for different failure types
- Recommend retraining when model performance is at risk

The objective is not only to build an accurate predictive model, but to design a robust end-to-end MLOps pipeline that supports real-world deployment, monitoring, and decision-making.





Previous

MLOps Capstone - Introduction

Next

Files

